

RESOURCE-NORMALIZED SCALING LAWS FOR BFT CONSENSUS

Sizing a permissioned BFT validator set for e-governance trades falling throughput against rising fault-detection power as n grows. We propose resource-normalized measurement and a bi-factor model; in the unsaturated regime with zero-mean noise the single-factor exponent tends to $\gamma\lambda - \beta$ (an omitted-variable bias). On commodity CI for Besu/QBFT at a fixed quota $\beta = 1.651 \pm 0.151$; with a near-chance detector the detection constraint is relaxed (not enforced) and throughput alone sets the window [4, 4].

Problem statement. Permissioned Byzantine fault-tolerant (BFT) consensus underpins e-governance systems and poses a procurement question: how many validators n should the network have? Throughput favours small n : every validator executes every transaction, and PBFT-family protocols exchange $O(n^2)$ messages per block [1]. Fault-detection reliability favours large n . At the design stage, dedicated multi-host testbeds are often unavailable, while commodity CI runners co-locate many validators on few cores. Without normalization, single-factor fits $\text{TPS}(n) \propto n^\alpha$ confound consensus cost with host-resource contention and with the load generator's concurrency schedule. Published exponents then diverge in magnitude and sometimes in sign; previous calibrations, extrapolated to sixty validators, differ by about an order of magnitude [2, 3].

The following tasks are addressed: separate the algorithmic cost of consensus from host-resource contention; explain the non-identifiability of the single-factor exponent along a concurrency path; formulate a chance-constrained validator-set sizing problem with throughput and detection constraints; calibrate the workflow on a reproducible commodity-CI campaign.

Essence of the study. A resource-normalized measurement (RNM) protocol is introduced: each validator container receives a fixed CPU quota q independent of n , and normalized throughput equals TPS / q . Under RNM the bi-factor scaling characterization takes the form

$$\text{TPS}(n, c) = T_0 \cdot g(c) \cdot n^{-\beta}, \quad g(c) = c^\gamma / [1 + (c/c^*)^\theta].$$

where c is client concurrency, $\beta > 0$ is the replica-count decay exponent, $\gamma \in (0, 1]$ is the concurrency exponent, and c^* is the saturation threshold (a USL-style form [4]). In the unsaturated regime $c \ll c^*$ the relation is log-linear. If measurements follow the path $\log c = \lambda \log n + \mu$ and the noise has zero mean and finite variance, the OLS slope of the single-factor model satisfies $\hat{\alpha} \xrightarrow{p} \gamma\lambda - \beta$; this is the standard omitted-variable bias [5]. In particular $\hat{\alpha} = -\beta$ only under a factorial design $\lambda = 0$; without reporting λ , published single-factor exponents should not be treated as transferable protocol properties across testbeds.

The calibrated exponent is the input to a chance-constrained program [6]: minimize a nondecreasing $\text{Cost}(n)$ subject to the lower prediction-interval bound on throughput meeting peak demand and the probability of missing a coordinated transaction-omission fault (an engineering Hoeffding-type bound [7]) not exceeding a prescribed level. Admissible sizes are discrete (efficient BFT sets have the form $3f + 1$), and the feasible set is a closed interval $[n_{\min}, n_{\max}]$; an empty interval certifies that no validator-set size meets both requirements, so the architecture or hardware class must change, not merely n .

The campaign ran in a public CI workflow on a four-core host: a Hyperledger Besu/QBFT network under the RNM protocol, 94 successful records, $n \leq 16$. Joint OLS at the reference quota $q = 0.25$ yields $\beta = 1.651 \pm 0.151$ and $\gamma = 0.178 \pm 0.070$ ($R^2 = 0.727$). Resampling along $\lambda = 1$ gives $\hat{\alpha} = -1.362$ versus the analytic $\gamma\lambda - \beta = -1.473$. Invariance of β across $q \in \{0.20, 0.25, 0.33\}$ is rejected ($p = 0.002$): at $q = 0.33$ throughput no longer grows with the quota and the total reservation exceeds host capacity. RNM therefore stabilizes β at a fixed quota, and coefficients are reported at $q = 0.25$. The omission/duplication detector has $\text{AUC} = 0.506$ (near chance); with it the detection constraint is unsatisfiable (it would require $n_{\min} \approx 3.3 \cdot 10^3$), so the formal feasible set is empty; the constraint is relaxed (not enforced) and throughput alone sets the window. The headline fit uses all n ; a separate refit on $n \leq 9$ only ($\beta = 1.300$) gives intervals covering 20 and 21 of 22 hold-out runs (90.9% / 95.5%) at $n \in \{11, 13, 16\}$. The constant β averages a slope that steepens with n (local values $\approx 1.4 \rightarrow 2.8$), which we attribute to residual host

contention as the reservation nq approaches the four host cores (RNM removes contention only while $nq < \text{host cores}$). The full-range β is conservative for $n_{\max}$ but optimistic for κ (13.0 vs. 16.7 under the small- n fit). A scenario of $\approx 3 \cdot 10^7$ ballots in 12 h gives a mean of ≈ 694 TPS; with an assumed peak-to-mean factor of ≈ 3 the peak is $\approx 2.1 \cdot 10^3$ TPS. Fault-free CI runs commit at most ≈ 89 TPS (at $n = 7$, $c = 16$, $q = 0.25$), so the model works in units of TPS/ q , and the scale $\kappa = 13.0$ is read as cores per validator on the target hardware. Linearity of throughput in the quota is the key assumption and the main extrapolation risk: κ is ≈ 52 times the calibrated quota, whereas throughput already stops growing at $q = 0.33$; κ is a requirement to verify, not a measured capability. The window then collapses to $[4, 4]$. Doubled demand or a 200 ms emulated WAN RTT (β rises from 1.459 to 2.708 on the WAN-experiment grid) at fixed κ empties the feasible set; $\kappa \approx 26$ or ≈ 73 , respectively, would be required.

Conclusions. Divergence of single-factor BFT scaling exponents can arise from the measurement design (the concurrency path λ), not only from protocol properties. Resource-normalized calibration at $\lambda = 0$ stabilizes the consensus exponent at a fixed quota and turns it into a reproducible validator-set sizing workflow; the throughput–detection trade-off is formulated but not demonstrated with the measured detector. For the studied Besu/QBFT deployment on commodity CI, the procurement answer is the smallest admissible set $n = 4$; larger n degrades predicted throughput. Numerical coefficients are deployment-specific; the methodology is intended for recalibration under matching measurement conditions. Code, data and CI-campaign artifacts: https://github.com/bpenyak/Resource_Normalized_Scaling_Laws_for_Byzantine_Consensus.

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